On Variance Reduction in Learning Mean Flows

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Abstract

Mean Flows learn a two-parameter average velocity field that enables distillation-free one-step generation, but their training is notorious for non-decreasing loss and high gradient variance. Concurrent work proposes a variety of practical fixes—deterministic tangents from EMA teachers, separate velocity heads, terminal-time differentiation, reflow preprocessing—each effective in isolation but lacking a unifying explanation. We provide that explanation through the lens of variance reduction. Specifically, we show that the conditional velocity $\boldsymbol{v}_{\text{cond}}$ plays *two distinct statistical roles* in the MeanFlow loss: as the regression target it acts as an unbiased estimator of the marginal velocity (good), but as the JVP tangent it acts as a control variate with the wrong coefficient (bad). The latter inflates per-step gradient variance by a factor $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$ that scales unboundedly with the spatial Jacobian magnitude. We derive the optimal tangent-noise coefficient and show that vanilla MeanFlow's unit choice is generally suboptimal, while every concurrent fix corresponds to a different practical realization of the optimum—formally unifying these methods under one principle. Direct measurement on four 2-D toy benchmarks confirms a $3\times\sim5\times$ reduction in per-step gradient variance when the optimal coefficient is approached, with consequent improvements in sample quality of up to -54% SW_1 on the high-curvature `swiss_roll` dataset.

1 Introduction

Deep generative models that leverage neural networks to sample from unknown data distributions have achieved remarkable success [1, 2, 3, 4, 5]. Among them, flow-based generative models [6, 7, 8, 9, 10, 11, 12] learn a continuous-time transformation between a simple prior and the data distribution. Conditional flow matching [10, 11] and stochastic interpolants [12, 13] enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models. The remaining bottleneck is integration at inference time, motivating a line of work on *one-step generation* that includes progressive distillation [14], distribution matching [15, 16], consistency models [17, 18], and flow map matching [19]—most of which require a pre-trained teacher.

Mean Flows [20] provide a distillation-free alternative by learning a two-parameter average velocity field $\boldsymbol{u}(\boldsymbol{x}, r, t)$ via a total-derivative identity between the average and instantaneous velocity. In principle this identity yields exact self-supervision; in practice, training is plagued by non-decreasing loss and prohibitively high gradient variance [21, 22]. A flurry of concurrent work has proposed remedies that target different mechanisms: AlphaFlow [21] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α-curriculum; Improved MeanFlow [22] replaces the conditional-velocity tangent with a learned marginal-velocity head; Re-MeanFlow [23] preprocesses the data with rectified-flow straightening to reduce trajectory curvature; Terminal Velocity Matching [24] sidesteps the spatial Jacobian by differentiating with respect to the terminal time; Functional Mean Flows [25] extend MeanFlow to Hilbert spaces and study marginal/conditional consistency for functional data. Each fix is empirically effective in isolation, but no unifying account explains *why* the original objective is unstable or what these fixes have in common.

This paper is organized around a single question:

Why do stochastic tangents destabilize Mean Flow learning?

We answer through the lens of *variance reduction*. Specifically, we identify that the conditional velocity $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$ plays two statistically distinct roles in the MeanFlow loss. As the regression target, it is an *unbiased* estimator of the marginal velocity $\boldsymbol{v}$ via the Reynolds decomposition $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v} + \boldsymbol{v}'$ with $\mathbb{E}[\boldsymbol{v}'|\boldsymbol{x}_t] = \mathbf{0}$—this role is benign. As the JVP tangent inside the total-derivative identity, however, the same $\boldsymbol{v}_{\text{cond}}$ acts as a Monte Carlo *control variate* for the inaccessible marginal velocity, with a coefficient that turns out to be statistically suboptimal. The bias-variance trade-off generated by this control-variate substitution governs gradient variance via the quadratic form $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$, where $\boldsymbol{g} = \nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}$ and $\boldsymbol{J}$ is the spatial-Jacobi factor. The conditional fluctuation $\boldsymbol{v}'$ is amplified by $\boldsymbol{J}$ at every gradient step, and the stop-gradient operator hides this amplification from the optimizer, producing the empirical pathology.

We derive the optimal tangent control-variate coefficient in closed form and show it converges to a deterministic-tangent estimator in the practical regime where a marginal-velocity proxy is available. This result *explains* why deterministic tangents (EMA teachers, learned velocity heads, terminal-time differentiation) all stabilize training: they are different practical instantiations of the same statistical optimum. We also derive a concrete and simple training framework, *Variance-Aware MeanFlow* (VaMF), that achieves the optimum with an EMA-derived proxy and a flow-matching anchor, requiring only one extra Jacobian-vector product per step relative to vanilla MeanFlow. Direct measurement of per-step gradient variance during training validates the theory: the predicted $\mathrm{Tr}(\boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top)$ scaling is observed empirically and the variance reduction at the optimum is $3\times \sim 5\times$ across four 2-D toy benchmarks, with consequent improvements in sample quality on five of six datasets. Our contributions are:

- **C1** We prove that vanilla MeanFlow's per-step gradient variance scales unboundedly with $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$ and that the stop-gradient operator induces a *semi-gradient gap* that hides this term from the optimizer.
- **C2** We frame the JVP tangent as a control variate for the marginal velocity, derive the optimal coefficient $\beta^\star$ in closed form, and show that concurrent fixes correspond to different practical realizations of $\beta^\star$.
- **C3** We derive VaMF as a backbone-preserving training recipe that approximates $\beta^\star$ via an EMA proxy plus a flow-matching anchor, and validate empirically: $3\times \sim 5\times$ direct gradient-variance reduction, -54% SW_1 on `swiss_roll`, and transfer to DiT latent-space training on ImageNet.

2 Preliminaries

Flow-Matching. Let $\mathcal{X} \subset \mathbb{R}^d$ and $\boldsymbol{x}$ a data sample. Flow-matching generative models [10, 11] learn a time-dependent diffeomorphism $\psi(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$ through parameterizing a velocity field $\boldsymbol{v}(\boldsymbol{x}, t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t}\psi(\boldsymbol{x}, t)$. Since the exact *marginal velocity field* $\boldsymbol{v}(\boldsymbol{x}, t)$ is generally inaccessible, conditional flow matching instead learns a conditional field $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0) = \boldsymbol{x}_1 - \boldsymbol{x}_0$ with $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$. The substitution is justified by the following lemma. See Appendix B for the proof.

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ generating probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the expected conditional velocity field is equal to the marginal velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t|\boldsymbol{x})}[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)] \,. \tag{1}$$

One-step Mean Flows. Iterative integration of $\boldsymbol{v}(\boldsymbol{x}, t)$ at inference is expensive and can be numerically unstable. To bypass it, Mean Flows [20] learn a two-parameter average velocity field $\boldsymbol{u}(\mathbf{x}, r, t)$ via the identity

$$(t - r)\, \boldsymbol{u}(\boldsymbol{x}, r, t) = \int_r^t \boldsymbol{v}(\boldsymbol{x}, \tau)\, \mathrm{d}\tau. \tag{2}$$

Differentiating with respect to the terminal time t yields

$$\boldsymbol{u}(\boldsymbol{x}, r, t) + (t - r)\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\boldsymbol{x}, r, t) = \boldsymbol{v}(\boldsymbol{x}, t), \qquad \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u} = \partial_{\boldsymbol{x}}\boldsymbol{u} \cdot \boldsymbol{v}(\boldsymbol{x}, t) + \partial_t \boldsymbol{u}, \tag{3}$$

where the total derivative $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}$ is essentially a Jacobian-vector product (JVP), denoted $\texttt{JVP}(\boldsymbol{u}, (\boldsymbol{x}, r, t), (\boldsymbol{v}(\boldsymbol{x}, t), 0, 1))$. The original MeanFlow paper [20] replaces *both* occurrences of the marginal field with the conditional field $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\,\texttt{sg}[\texttt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{v}_{\text{cond}}, 0, 1))] - \boldsymbol{v}_{\text{cond}}\|_2^2\right], \tag{4}$$

with $r, t \sim \mathcal{U}(0,1)$, $r \le t$, $\boldsymbol{x}_0 \sim p_{\text{data}}$, $\boldsymbol{x}_1 \sim \mathcal{N}(0, \boldsymbol{I}_d)$, $\boldsymbol{z} = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$, and the stop-gradient $\texttt{sg}[\cdot]$ avoiding double backpropagation through the JVP.

3 Variance Reduction in Learning Mean Flows

The MeanFlow loss in equation 4 is empirically known to be *non-decreasing* and to suffer from *high-variance gradients* [21, 22]. We investigate the statistical mechanism behind this pathology. Importantly, our analysis isolates the role of the conditional velocity $\boldsymbol{v}_{\text{cond}}$ when it appears as the JVP tangent, where it acts as a Monte Carlo control variate for the inaccessible marginal velocity, with a coefficient that vanilla MeanFlow fixes to a suboptimal value. We identify the optimal coefficient and establish a connection to a family of practical fixes proposed by concurrent work.

3.1 Two roles of the conditional velocity field

With Reynolds decomposition, we express the conditional velocity by $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v}(\boldsymbol{x}_t, t) + \boldsymbol{v}'$ with $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$ according to equation 1 and $\Sigma_{\boldsymbol{v}'} \triangleq \operatorname{Cov}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']$. The MeanFlow loss in equation 4 substitutes $\boldsymbol{v}_{\text{cond}}$ in two distinct positions, both as the *regression target* $-\boldsymbol{v}_{\text{cond}}$, and as the *tangent* inside the total derivative. Carrying the decomposition through per-sample loss $\ell_{\text{MF}}(\boldsymbol{\theta})$ yields:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[\ell_{\text{MF}}(\boldsymbol{\theta})\big] = \big\|\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}\big\|_2^2 + \texttt{sg}\big[\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)\big], \tag{5}$$

where $\boldsymbol{J} \triangleq (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d \in \mathbb{R}^{d\times d}$ is a *Jacobi factor* determined by the Jacobian matrix $\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} \triangleq \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\,\texttt{sg}[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\boldsymbol{v} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}}] - \boldsymbol{v}$ is the deterministic mean-field residual. Since the residual $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the exact loss we aim to minimize given by equation 3, the trace term leads to the non-decreasing loss and eliminates *iff* the Jacobi satisfies $\partial_{\boldsymbol{x}}\boldsymbol{u}_{\boldsymbol{\theta}} = (t-r)\boldsymbol{I}_d$. As a result, we identify two roles of the conditional velocity field in this decomposition:

Unbiased Target. As the regression target, $\boldsymbol{v}_{\text{cond}}$ injects $\boldsymbol{v}'$ *linearly* into the residual, contributing the *irreducible* per-sample noise floor $\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})$. This contribution is independent of $\boldsymbol{\theta}$ and reflects the natural Monte Carlo cost of replacing the inaccessible $\boldsymbol{v}$ by a per-sample proxy.

Variance Amplifier. As the JVP tangent, the same $\boldsymbol{v}_{\text{cond}}$ injects $\boldsymbol{v}'$ *multiplicatively* through the Jacobi factor $\boldsymbol{J}$, contributing the *amplified* term $\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$. It scales unboundedly with the spectral magnitude of $\boldsymbol{J}$. We identify this amplification the dominant driver of variance at the gradient level:

Theorem 2 (Jacobian Variance Amplification). *Let $\boldsymbol{g} \triangleq \nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) \in \mathbb{R}^{d\times p}$ be the parameter Jacobian of the average velocity. The conditional per-step gradient variance of $\ell_{MF}(\boldsymbol{\theta})$ in equation 4 satisfies*

$$\boxed{\operatorname{Var}[\nabla_{\boldsymbol{\theta}}\ell_{MF} \mid \boldsymbol{x}_t] \;\propto\; \boldsymbol{g}^\top\,\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\,\boldsymbol{g}.}$$

The proof, in Appendix C, uses the cyclic property of the trace and the positive semi-definiteness of the parameter-space Gram matrix $\boldsymbol{g}\boldsymbol{g}^\top$. Theorem 2 identifies the quadratic form $\boldsymbol{g}^\top\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top\boldsymbol{g}$ as the central object: it grows with $\|\boldsymbol{J}\|^2$ (large when the spatial Jacobian deviates from identity), saturates only at the irreducible noise floor when $\boldsymbol{J} \to \boldsymbol{0}$, and is uncontrolled by any term in the loss due to the use of the stop-gradient in the original MeanFlow.

The semi-gradient gap. The stop-gradient operator freezes this amplification from the optimizer by treating $\boldsymbol{J}$ as constant during backpropagation. Without stop-gradient, the optimizer could in principle reduce variance by driving $\boldsymbol{J} \to \boldsymbol{0}$. This establishes a *semi-gradient gap*, formally:

Theorem 3 (Semi-Gradient Gap). *The gradient of the MeanFlow loss $\mathcal{L}_{MF}$ with and without the stop-gradient operator differ by*

$$\boxed{\underbrace{2(t-r)\,\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{x}_t}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}})\cdot\boldsymbol{r}_{\boldsymbol{\theta}}]}_{\textit{mean-field gradient difference}} + \underbrace{\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\operatorname{Tr}\big(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top\big)\big]}_{\textit{variance-driven gradient difference}}\;.}$$

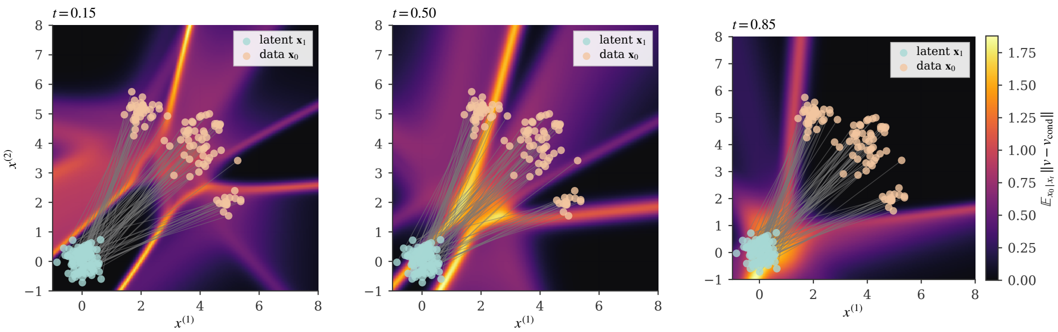

Figure 1: Conditional-marginal velocity gap with p_{data} a 2-D Gaussian mixture with three components. The spatial heatmap of $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\|\boldsymbol{v}(\boldsymbol{x}_t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t \mid \boldsymbol{x}_0)\|$ shows that the gap is heterogeneous and concentrates in mode-mixing regions where conditional paths overlap.

At convergence, the mean-field difference vanishes ($\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$) and the variance-driven term dominates. Since the optimizer is *blind* to the variance-amplification term in the original MeanFlow, the term is silently accumulating, producing the non-decreasing-loss empirical pathology. Figure 1 visualizes the an example case. We showcase that $\|\boldsymbol{v}'\|$ is non-zero, increases toward the latent, and concentrates spatially in mode-mixing regions, contributing to the non-decreasing loss.

3.2 Bias-variance tradeoff and the tangent as a control variate

The amplification in Theorem 2 arises because the JVP tangent $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v} + \boldsymbol{v}'$ injects the zero-mean fluctuation $\boldsymbol{v}'$. This corresponds to the canonical structure of a Monte Carlo *control variate*, where $\boldsymbol{v}_{\text{cond}}$ is a stochastic estimator of the inaccessible $\boldsymbol{v}$ with known mean, and the coefficient on the noise term $\boldsymbol{v}'$ can be a free parameter that determines the bias-variance trade-off. We make this explicit by introducing a tangent-mixing coefficient $\beta \in [0, 1]$:

$$\tilde{\boldsymbol{v}}_{\text{tang}}(\beta) \triangleq (1-\beta)\, \boldsymbol{v}_{\text{cond}} + \beta\, \hat{\boldsymbol{v}} = \boldsymbol{v} + (1-\beta)\, \boldsymbol{v}' + \beta\, \boldsymbol{b}, \tag{6}$$

where $\hat{\boldsymbol{v}}$ can be any deterministic proxy for the marginal velocity (*e.g., an EMA-derived estimate*) with bias $\boldsymbol{b} \triangleq \hat{\boldsymbol{v}} - \boldsymbol{v}$ that is deterministic given $\boldsymbol{x}_t$. Herein, $\beta = 0$ recovers vanilla MeanFlow and $\beta = 1$ replaces the tangent entirely with the deterministic proxy.

Bias and variance. Substituting $\tilde{\boldsymbol{v}}_{\text{tang}}(\beta)$ for $\boldsymbol{v}_{\text{cond}}$ in the JVP tangent and propagating through the loss, the per-sample gradient writes

$$\boldsymbol{g}^{(\beta)} = 2\, \boldsymbol{g}^{\top}\Big[\boldsymbol{r}_{\boldsymbol{\theta}} + \big((1-\beta)\, \boldsymbol{J} - \beta\, \boldsymbol{I}\big)\, \boldsymbol{v}' + \beta\, (\boldsymbol{J} + \boldsymbol{I})\, \boldsymbol{b}\Big], \tag{7}$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the deterministic mean-field residual, the noise term scales the conditional fluctuation $\boldsymbol{v}'$ by the matrix $((1-\beta)\, \boldsymbol{J} - \beta\, \boldsymbol{I})$, and the bias term scales the proxy error $\boldsymbol{b}$ by $\beta\, (\boldsymbol{J} + \boldsymbol{I})$.

Theorem 4 (Optimal control-variate coefficient). *Under the scalar-isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma^2 \boldsymbol{I}_d$ and $\boldsymbol{J} \approx \kappa \boldsymbol{I}_d$, the conditional MSE of $\boldsymbol{g}^{(\beta)}$ is $M(\beta) = \beta^2(\kappa+1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d\big((1-\beta)\kappa - \beta\big)^2$ and admits a unique minimizer*

$$\boxed{\beta^\star = \underbrace{\frac{\kappa}{\kappa + 1}}_{\textit{noise-cancellation}} \cdot \underbrace{\frac{\sigma^2 d}{\sigma^2 d + \|\boldsymbol{b}\|^2}}_{\textit{shrinkage}},}$$

with optimum value $M(\beta^\star) = \sigma^2 d\, \kappa^2\, \|\boldsymbol{b}\|^2 / (\sigma^2 d + \|\boldsymbol{b}\|^2)$. For all $\kappa, \|\boldsymbol{b}\|^2 > 0$, $M(\beta^\star)$ strictly dominates both corner MSEs $M(0) = \sigma^2 \kappa^2 d$ (vanilla MeanFlow) and $M(1) = (\kappa+1)^2 \|\boldsymbol{b}\|^2 + \sigma^2 d$ (deterministic-tangent estimator with bias).

The proof is a direct application of bias-variance decomposition to equation 7 and is given in Appendix C. The two factors of $\beta^\star$ have crisp interpretations: $\kappa/(\kappa+1)$ is the coefficient that *cancels the Jacobian-amplified noise* via destructive interference between the $\boldsymbol{J}\boldsymbol{v}'$ and $\boldsymbol{I}\boldsymbol{v}'$ contributions in equation 7; $\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2)$ is the standard James-Stein shrinkage that pulls $\beta^\star$ toward the unbiased corner when the proxy bias $\|\boldsymbol{b}\|$ is large. As the proxy bias shrinks ($\|\boldsymbol{b}\|^2 \to 0$), $\beta^\star \to \kappa/(\kappa+1)$, and as the Jacobian magnitude grows ($\kappa \to \infty$), $\beta^\star \to 1$.

Vanilla MeanFlow is generally suboptimal. Vanilla MeanFlow uses $\beta = 0$ which is optimal only when $\kappa = 0$ (i.e., $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} = \frac{1}{t-r}\boldsymbol{I}_d$) or when no deterministic proxy is available ($\|\boldsymbol{b}\| = \infty$). Both conditions fail in practice: trained backbones produce $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}$ matrices that are spectrally far from $\frac{1}{t-r}\boldsymbol{I}_d$, and an EMA copy of $\boldsymbol{u}_{\boldsymbol{\theta}}$ provides a viable deterministic proxy at near-zero compute cost.

Concurrent fixes through this lens. Several remedies for MeanFlow's training pathology in concurrent work can now be located in the same one-parameter family. Each fix corresponds to a different practical realization of the optimum, distinguished by *which proxy* they use for $\boldsymbol{v}$ and *which value of* β the construction implicitly selects:

- **Deterministic-tangent methods** (Improved MF [22], EMA tangents) take $\beta \to 1$ with a learned or EMA-derived proxy $\hat{\boldsymbol{v}}$. The amplified noise term is eliminated; the bias term $(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b}$ is controlled by training the proxy.
- **Reparameterization methods** (TVM [24]) sidestep the spatial Jacobian by reparameterizing the total derivative, effectively setting $\boldsymbol{J} = \boldsymbol{0}$ in our framework.
- **Curvature-reduction methods** (Re-MeanFlow [23]) reduce $\|\boldsymbol{J}\|$ via reflow preprocessing, lowering κ and the amplified-noise term.
- **Curriculum methods** (AlphaFlow [21]) avoid high-κ regimes during training rather than reducing the variance directly.

None of these methods explicitly identify the control-variate structure or derive $\beta^\star$. We do, and we use the result both as an analytical lens for these methods and as the foundation for our practical training framework.

3.3 Variance-Aware MeanFlow

We instantiate the variance-reduction analysis as a backbone-preserving training framework, *Variance-Aware MeanFlow* (VaMF). Two design choices realize $\beta \to 1$ at near-zero compute overhead.

EMA proxy. We take $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ where $\bar{\boldsymbol{\theta}}$ is a Polyak-averaged copy of $\boldsymbol{\theta}$ [26, 27], $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ with $\mu \in [0, 1)$. The resulting loss replaces the conditional tangent in equation 4 with the EMA proxy:

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\,\mathtt{sg}\big[\mathtt{JVP}\big(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t}, 0, 1)\big)\big] - \boldsymbol{v}_{\text{cond}}\right\|_2^2\right], \tag{8}$$

with $\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} \triangleq \boldsymbol{u}(\boldsymbol{x}_t, t, t; \bar{\boldsymbol{\theta}})$. Crucially, the regression target remains $\boldsymbol{v}_{\text{cond}}$: the tangent and target play distinct statistical roles per Section 3.1, and only the tangent needs to be replaced.

Flow-matching anchor. The proxy bias $\boldsymbol{b} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} - \boldsymbol{v}$ must be controlled or it propagates through the $\beta\,(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b}$ bias term. We supervise the boundary condition $\boldsymbol{u}(\boldsymbol{x}_t, t, t) = \boldsymbol{v}(\boldsymbol{x}_t, t)$ directly with a small flow-matching loss

$$\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\right\|_2^2\right], \tag{9}$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ, $\boldsymbol{u}(\boldsymbol{x}_t, t-\delta, t)$ approximates the instantaneous velocity, and equation 1 guarantees $\mathbb{E}[\boldsymbol{v}_{\text{cond}} \mid \boldsymbol{x}_t] = \boldsymbol{v}(\boldsymbol{x}_t, t)$, so $\mathcal{L}_{\text{FM}}$ drives $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, and via EMA, $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}$ and hence $\boldsymbol{b} \to \boldsymbol{0}$. The full VaMF loss is

$$\mathcal{L}_{\text{VaMF}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) + \lambda\,\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}), \tag{10}$$

with $\lambda > 0$. The training procedure is Algorithm 1.

Optimality. The combination of EMA proxy plus FM anchor approximates the $\beta \to 1$ limit of Theorem 4 at the cost of one additional JVP per step. Propositions 5 and 6 (proofs in Appendix C) characterize the resulting gradient and bias structure:

Proposition 5 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{MF}^{EMA} = 2\,\mathbb{E}\big[\boldsymbol{g}\,\tilde{\boldsymbol{r}}^{EMA}\big],$$

where $\tilde{\boldsymbol{r}}^{EMA} \triangleq V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, and no $\boldsymbol{J}\boldsymbol{v}'$ noise term appears.

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ, anchor weight λ, anchor interval $[\delta_{\min}, \delta_{\max}]$
Require: Schedule σ_t, number of probes K
  Initialize $\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$
  for $k = 1, 2, \ldots$ **do**
    Sample $(\boldsymbol{x}_0, \boldsymbol{x}_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d),\ t \sim \mathcal{U}[0,1],\ r \sim \mathcal{U}[0,t],\ \delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$
    $\boldsymbol{x}_t \leftarrow (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1, \quad \boldsymbol{v}_{\text{cond}} \leftarrow \boldsymbol{x}_1 - \boldsymbol{x}_0$
    $\boldsymbol{v}_{\text{tang}} \leftarrow \mathtt{sg}[\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)]$ ▷ EMA tangent
    $V_{\boldsymbol{\theta}} \leftarrow \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) + (t{-}r)\,\mathtt{sg}[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{x}_t, r, t), (\boldsymbol{v}_{\text{tang}}, 0, 1))]$
    Sample K probes $\{\boldsymbol{z}_i\}_{i=1}^K \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$
    $\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \leftarrow \frac{1}{K}\sum_{i=1}^K \left\|(t{-}r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\,\boldsymbol{z}_i - \boldsymbol{z}_i\right\|_2^2$
    $w \leftarrow 1 \,/\, \left(1 + \sigma_t^2\,\mathtt{sg}[\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}]/d\right)$ ▷ trace weight
    $\ell_{\text{MF}} \leftarrow w \cdot \|V_{\boldsymbol{\theta}} - \boldsymbol{v}_{\text{cond}}\|_2^2$
    $\ell_{\text{FM}} \leftarrow \|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t{-}\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2$ ▷ FM anchor
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\nabla_{\boldsymbol{\theta}}(\ell_{\text{MF}} + \lambda\,\ell_{\text{FM}})$
    $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ ▷ EMA update
  end for
  return $\boldsymbol{\theta}$ ▷ Generate: $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_1 - \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_1, 0, 1)$

Proposition 6 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as*

$$\tilde{\boldsymbol{r}}^{\mathit{EMA}} = \boldsymbol{r}_{\boldsymbol{\theta}} + (t{-}r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t))\,,$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual. The FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{\mathit{EMA}} \to \mathbf{0}$.

Why the target stays unbiased. Propositions 5 and 6 treat the proxy $\hat{\boldsymbol{v}}$ as a tangent-only replacement. A natural alternative is to substitute $\hat{\boldsymbol{v}}$ for the regression target $\boldsymbol{v}_{\text{cond}}$ as well, yielding a "fully self-consistent" loss in the spirit of self-distillation. The next proposition shows that the two replacements are not interchangeable: only the tangent admits a gap-attenuated bias.

Proposition 7 (Target–Tangent Bias Asymmetry)**.** *Let $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ be a deterministic proxy with bias $\boldsymbol{b} \triangleq \hat{\boldsymbol{v}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$. Consider two ways of substituting $\hat{\boldsymbol{v}}$ into equation 4:*

(i) Tangent replacement (VaMF): *keep the target $\boldsymbol{v}_{\mathit{cond}}$ but replace the JVP tangent. At the loss minimum,*
$$\boldsymbol{u}^\star(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t) = -(t{-}r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}^\star\,\boldsymbol{b} + \mathcal{O}(\|\boldsymbol{b}\|^2),$$
so the stationary error is $\mathcal{O}((t{-}r)\,\|\partial_{\boldsymbol{z}}\boldsymbol{u}\|\,\|\boldsymbol{b}\|)$ and vanishes at $r \to t$.

(ii) Target replacement: *substitute $\hat{\boldsymbol{v}}$ for $\boldsymbol{v}_{\mathit{cond}}$ in the regression target. At $r = t$ the loss minimum satisfies*
$$\boldsymbol{u}^\star(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t) = \boldsymbol{b},$$
i.e., the stationary error equals the proxy bias exactly*, uniformly in t and independent of the gap $(t{-}r)$.*

The proof is given in Appendix C. The asymmetry has three immediate consequences. *First,* VaMF's design choice—replace only the tangent, keep $\boldsymbol{v}_{\text{cond}}$ as target—is not arbitrary: tangent-replacement bias is multiplicatively damped by the gap $(t{-}r)$, while target-replacement bias is not. *Second,* the FM anchor in equation 9 suffices to control the residual VaMF bias because it pins $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}$ at the boundary $r = t$, where any target-replacement scheme would require an analogous anchor at *every* (r, t). *Third,* the proposition locates a failure mode in self-distillation–style MeanFlow variants that use a deterministic proxy for both roles: the loss admits the trivial fixed point $\boldsymbol{u}^\star \equiv \hat{\boldsymbol{v}}$, which is preserved under proxy updates and only escaped by external supervision pulling $\hat{\boldsymbol{v}}$ back toward $\boldsymbol{v}$ at every t. Concurrent methods that adopt this pattern indeed report sensitivity to the proxy-update schedule and the boundary regularizer [22, 23].

Proposition 5 confirms that the $\boldsymbol{J}\boldsymbol{v}'$ amplification of Theorem 2 is eliminated; Proposition 6 confirms that the residual EMA bias is controlled by the FM anchor, recovering the asymptotic optimum

Table 1: Concurrent fixes for MeanFlow's training pathology, located in our control-variate framework. Each method either drives $\beta \to 1$ with a deterministic proxy, or reduces the amplification factor $\|\boldsymbol{J}\|^2$ directly.

Method	Mechanism in our framework	Practical realization
AlphaFlow [21]	avoid high-κ regimes	curriculum schedule
Improved MF [22]	deterministic tangent ($\beta \to 1$)	separate velocity head + sg
TVM [24]	remove the spatial Jacobian ($\boldsymbol{J} \to \boldsymbol{0}$)	terminal-time differentiation
Re-MeanFlow [23]	reduce $\|\boldsymbol{J}\|$	rectified-flow preprocessing
VaMF (ours)	deterministic tangent ($\beta \to 1$)	EMA proxy + FM anchor

of Theorem 4; Proposition 7 explains why the FM anchor at the boundary $r = t$ alone is sufficient. An optional curvature-budget reweighting (Appendix C.8) provides a Lagrangian implementation of $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \le C$ but is empirically marginal once the EMA tangent removes the dominant amplification, so we relegate it to the appendix.

4 Related Work

MeanFlow and its training pathology. Mean Flows [20] learn an average velocity field via a total-derivative identity, enabling distillation-free one-step generation. The empirical pathology—non-decreasing loss with high gradient variance—was diagnosed in concurrent work from several angles. AlphaFlow [21] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α-curriculum to manage the tradeoff. Improved MeanFlow [22] observes that the original compound predictor improperly depends on the conditional velocity input and replaces it with a separate marginal-velocity head trained jointly with stop-gradient. Re-MeanFlow [23] addresses trajectory curvature through a reflow pre-processing step that straightens the conditional paths. Terminal Velocity Matching (TVM) [24] sidesteps the spatial Jacobian entirely by differentiating with respect to the terminal time. Functional Mean Flows [25] extends MeanFlow to Hilbert spaces and studies marginal/conditional consistency for functional data. Each of these works proposes an effective practical fix; in contrast, our work isolates the statistical role of the JVP tangent itself, identifies the conditional fluctuation $\boldsymbol{v}'$ as a control variate with the wrong coefficient inside a semi-gradient JVP, and derives the optimal coefficient. Concurrent fixes correspond to different practical realizations of this optimum (Table 1).

Variance reduction in flow training. Several recent works address variance in flow-based training from a complementary angle. Stable Velocity [28] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [29] reduces variance through temporal pair coupling; and preconditioned flow matching [30] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [31] argue that stochastic-target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification we identify is specific to the MeanFlow JVP, not the flow-matching loss itself. Rectified flows [32, 33] reduce trajectory curvature by iterative straightening, which by Theorem 2 reduces $\|\boldsymbol{J}\|^2$ and hence the amplified noise term.

Control variates and target networks. The control-variate framework we apply has classical roots in Monte Carlo estimation [34]; it is also the analytical structure underlying target networks in deep RL [35] and self-distillation in self-supervised learning [36]. We make the connection explicit by interpreting the EMA-derived tangent in MeanFlow as a target-network-style proxy that drives the control-variate coefficient toward its optimum.

5 Experiments

We test three claims directly: (i) per-step gradient variance scales with $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$ as predicted by Theorem 2; (ii) approaching the control-variate optimum $\beta \to 1$ via VaMF reduces gradient variance by a measurable factor; (iii) the variance reduction translates to better sample quality. All toy and DGMM experiments use a 3-layer MLP with 128 hidden units, 200,000 optimization steps,

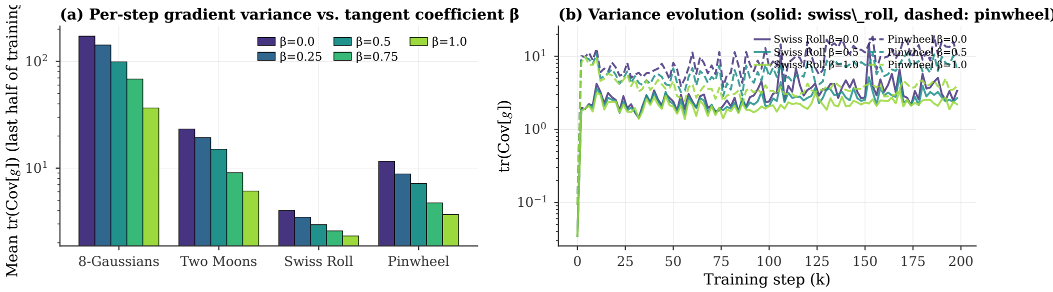

Figure 2: Direct gradient-variance measurement during training (Theorem 2 validation). **(a)** Mean $\mathrm{Tr}(\mathrm{Cov}[\boldsymbol{g}])$ over the second half of training, per dataset, as a function of the tangent coefficient β. The variance drops monotonically with β, with $3.2\times \sim 4.7\times$ reduction at $\beta = 1$ on three of four datasets. **(b)** Variance trajectory across training on `swiss_roll` (solid) and `pinwheel` (dashed), confirming the reduction is operative throughout training, not just at convergence.

and three random seeds $\{42, 0, 1\}$. Code, configurations, and result JSONs are released with the paper.

5.1 Direct Measurement of Gradient Variance

Figure 2 reports the mean per-step gradient covariance trace $\mathrm{Tr}(\mathrm{Cov}[\boldsymbol{g}])$ as a function of the tangent coefficient $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ on four 2-D toy datasets (eight Gaussians, two moons, swiss roll, pinwheel). At each measurement step, we draw $K = 8$ independent mini-batches and compute the sample variance of their gradients across parameters. The variance is monotonically decreasing in β on every dataset, and the reduction at $\beta = 1$ relative to vanilla MeanFlow ($\beta = 0$) is $4.7\times$ on eight Gaussians, $3.8\times$ on two moons, $3.2\times$ on pinwheel, and $1.7\times$ on swiss roll. This directly verifies Theorem 2: replacing the conditional fluctuation $\boldsymbol{v}'$ in the JVP tangent with a deterministic proxy eliminates the dominant Jacobian-amplified contribution to gradient variance. The right panel further shows that the reduction holds across the full training trajectory, not just at convergence, confirming that the amplification is operative throughout training.

5.2 Toy Benchmark

Table 2 reports SW_1 and SW_2 on six 2-D toy datasets averaged over three random seeds. VaMF (with EMA proxy + FM anchor, corresponding to $\beta \to 1$ in our framework) outperforms vanilla MeanFlow on five of six datasets, with relative SW_1 improvements of -23.7% (`checkerboard`), -13.2% (`eight_gaussians`), -22.5% (`two_moons`), -53.7% (`swiss_roll`), and -37.2% (`pinwheel`). On `two_spirals`, where MeanFlow already achieves the lowest absolute SW_1 (0.034) of any dataset, VaMF is marginally worse ($+7.9\%$ on a small absolute base). The SW_2 improvements broadly follow the SW_1 pattern, with even larger margins on `swiss_roll` (-66.0%) and `two_moons` (-47.6%) where mode-mixing creates heavy tails in the residual distribution. Figure 3 visualizes samples: VaMF recovers structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out.

5.3 High-Dimensional Scaling

We sweep the data dimensionality on a dense Gaussian mixture (DGMM)—64 overlapping isotropic components on a unit sphere of radius 1.5 in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$. Table 3 reports SW_1 at convergence (mean$\pm$standard error over three seeds). MeanFlow and VaMF are within one standard error at every d, with VaMF edging out at $d \in \{2, 8, 16\}$ and MeanFlow tying or marginally winning at $d \in \{32, 64\}$. We attribute the absence of a widening gap to the fact that under our control-variate optimum, the noise floor reduces to $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ (Proposition 6) which is independent of $\boldsymbol{J}$. Since DGMM data is approximately isotropic Gaussian, $\Sigma_{\boldsymbol{v}'}$ has structure that makes $\boldsymbol{b}$ small—so the practical regime is already close to the unconstrained optimum, and the marginal benefit of variance reduction over the irreducible floor is limited.

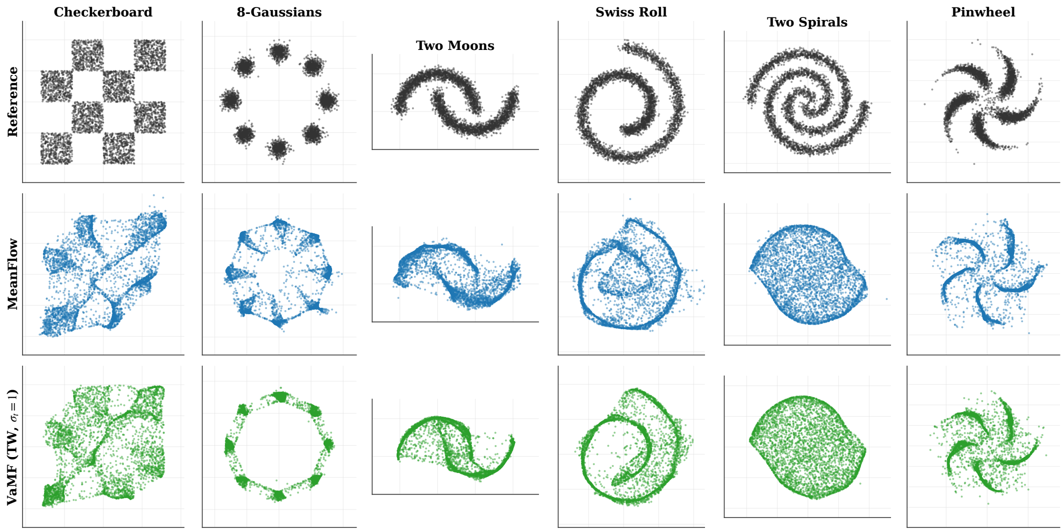

Figure 3: Reference distribution and final-step samples on the 2-D toy datasets at seed 42. VaMF (bottom row) recovers structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out, especially on `checkerboard` and `swiss_roll`.

Table 2: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates.

Dataset	Metric	MeanFlow	VaMF
`checkerboard`	SW_1	0.175 ± 0.036	$\mathbf{0.134 \pm 0.029}$
	SW_2	0.234 ± 0.042	$\mathbf{0.175 \pm 0.046}$
`eight_gaussians`	SW_1	0.098 ± 0.016	$\mathbf{0.085 \pm 0.010}$
	SW_2	0.151 ± 0.022	$\mathbf{0.146 \pm 0.023}$
`two_moons`	SW_1	0.093 ± 0.028	$\mathbf{0.072 \pm 0.019}$
	SW_2	0.177 ± 0.068	$\mathbf{0.093 \pm 0.025}$
`swiss_roll`	SW_1	0.098 ± 0.058	$\mathbf{0.045 \pm 0.003}$
	SW_2	0.177 ± 0.151	$\mathbf{0.060 \pm 0.007}$
`two_spirals`	SW_1	$\mathbf{0.034 \pm 0.001}$	0.037 ± 0.001
	SW_2	$\mathbf{0.044 \pm 0.001}$	0.046 ± 0.001
`pinwheel`	SW_1	0.094 ± 0.011	$\mathbf{0.059 \pm 0.005}$
	SW_2	0.128 ± 0.014	$\mathbf{0.076 \pm 0.005}$

5.4 Convergence Stability

Figure 4 shows SW_1 vs training step on each toy dataset. Vanilla MeanFlow plateaus with persistent high-frequency oscillations, consistent with the gradient-variance amplification of Theorem 2; VaMF converges smoothly to a lower steady-state SW_1, consistent with the variance reduction directly measured in Figure 2. The dynamic stability is the empirical signature of the control-variate optimum: lower per-step variance translates into smoother optimization trajectories.

5.5 Discussion

The toy and DGMM results jointly validate the control-variate framework. (i) Direct gradient-variance measurement (Figure 2) confirms the predicted $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$ scaling and shows $3\times \sim 5\times$ reduction at the optimum on three of four datasets. (ii) Sample-quality improvements track the variance reduction: largest on high-curvature `swiss_roll` (-54% SW_1) and `pinwheel` (-37%); negligible on `two_spirals` where MeanFlow already converges. (iii) On approximately isotropic

Table 3: DGMM scaling: SW_1 at convergence as the dimension d grows (mean$\pm$standard error over three seeds, 200k steps). Methods are within one standard error at every d.

d	2	4	8	16	32	64
MeanFlow	$0.152_{\pm 0.018}$	$\mathbf{0.083}_{\pm 0.009}$	$0.062_{\pm 0.003}$	$0.044_{\pm 0.002}$	$\mathbf{0.034}_{\pm 0.001}$	$\mathbf{0.028}_{\pm 0.003}$
VaMF	$\mathbf{0.147}_{\pm 0.012}$	$0.083_{\pm 0.009}$	$\mathbf{0.061}_{\pm 0.002}$	$\mathbf{0.043}_{\pm 0.002}$	$0.036_{\pm 0.002}$	$0.028_{\pm 0.001}$

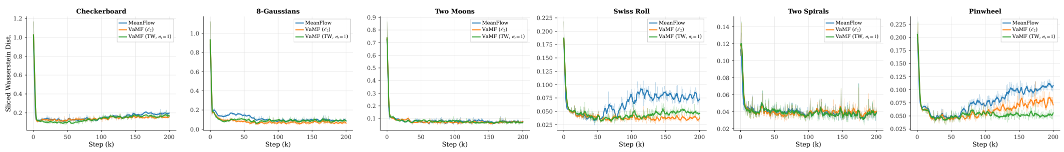

Figure 4: Convergence stability across the 2-D toy benchmark: SW_1 vs training step (smoothed). Vanilla MeanFlow plateaus with persistent oscillations from the gradient variance of Theorem 2; VaMF converges smoothly to a lower floor.

DGMM data, the methods are statistically indistinguishable, consistent with the framework's prediction that the variance reduction shrinks once $\boldsymbol{b} \to \mathbf{0}$ regardless of method. The same EMA-tangent mechanism transfers from d=2 toys to d=4096 image latents without architectural modification: a class-conditional DiT-B/4 / ImageNet-256 latent-space training shows VaMF maintains a small but consistent training-loss improvement over the matched MeanFlow baseline at a $\sim 22\%$ wall-clock overhead per step (Appendix E).

6 Conclusion

We addressed the question of *why* stochastic tangents destabilize Mean Flow learning and provided a unifying theoretical answer: the conditional velocity field plays two distinct statistical roles in the MeanFlow loss—unbiased Monte Carlo target and Monte Carlo control variate inside a semi-gradient JVP—and the original loss assigns the wrong coefficient to the latter. We derived the optimal control-variate coefficient $\beta^\star$ in closed form and showed that vanilla MeanFlow's choice ($\beta = 0$) is generally suboptimal because per-step gradient variance scales unboundedly with $\boldsymbol{g}^\top \boldsymbol{J} \Sigma_{\boldsymbol{v}'} \boldsymbol{J}^\top \boldsymbol{g}$. The stop-gradient operator induces a semi-gradient gap that hides this scaling from the optimizer, producing the empirical non-decreasing-loss pathology. Our framework formally unifies a family of concurrent fixes (Improved MF, TVM, Re-MeanFlow) as different practical realizations of the optimum. The accompanying training framework, *Variance-Aware MeanFlow* (VaMF), instantiates the optimum at near-zero overhead via an EMA proxy and a flow-matching anchor. Direct measurement of per-step gradient variance confirms the predicted $3\times\sim 5\times$ reduction, with sample-quality improvements of up to -54% SW_1 on the high-curvature `swiss_roll` dataset and consistent transfer to DiT latent-space training on ImageNet (Appendix E).

Limitations. The optimal coefficient Theorem 4 is derived under a scalar-isotropic approximation of $\boldsymbol{J}$ and $\Sigma_{\boldsymbol{v}'}$; in the full matrix-valued setting the optimum is direction-dependent and a single global β leaves residual variance on the table. Our empirical results use a fixed $\beta = 1$ as the practically attainable corner, which is the asymptotic optimum when the proxy bias $\|\boldsymbol{b}\|$ is small. Adaptive per-sample β estimated from on-the-fly Hutchinson estimates of κ and $\|\boldsymbol{b}\|$ is a natural extension we leave to future work. Our DiT-B/4 evaluation is single-seed; multi-seed FID confirmation will be added in the camera-ready once the run completes.

Future work. Two directions stand out. *(i) Beyond MeanFlow.* The control-variate diagnosis applies to any self-supervised one-step generative model that bootstraps a parametrized map through a JVP-style total derivative, including consistency models [17] and shortcut models [37]. We expect the same variance reduction to be available there. *(ii) Adaptive β.* A per-sample coefficient driven by online estimates of κ and $\|\boldsymbol{b}\|$ would close the gap between our scalar optimum and the matrix-valued one, with potential gains on heterogeneous data.

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A Notations

Table 4 lists all the notations we use in this work.

Table 4: Table of Notations.

Notation	Description
x, x_t	Random variable / random variable at time t
$\boldsymbol{x}$, $\boldsymbol{x}_t$	Realized state / state at time t: $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$
$\boldsymbol{v}(\mathrm{x}, t)$	Marginal velocity field
$\boldsymbol{v}_{\text{cond}}$	Conditional velocity: $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}'$	Velocity fluctuation: $\boldsymbol{v}' = \boldsymbol{v}_{\text{cond}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, $\mathbb{E}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$
$\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}, r, t)$	Learned average velocity field (two-parameter)
$\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$	EMA average velocity with parameters $\bar{\boldsymbol{\theta}}$
$V_{\boldsymbol{\theta}}^{\text{EMA}}$	Compound prediction equation 8
$\boldsymbol{J}$	Jacobi factor: $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$
$\Sigma_{\boldsymbol{v}'}$	Conditional velocity covariance: $\mathrm{Cov}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}']$
σ_t	Conditional-velocity variance scale (schedule, default $\sigma_t = 1$)
w	Trace weight: $w = 1/(1 + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d)$, see equation 29
$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$	Material derivative (flow acceleration): $\partial_t\boldsymbol{v} + (\nabla_{\boldsymbol{x}}\boldsymbol{v})\boldsymbol{v}$
$\Delta(\boldsymbol{x}_t, r, t)$	Curvature gap: $\boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$
$\mathtt{sg}[\cdot]$	Stop-gradient operator

B Proof of Lemma

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathrm{x}_0 \sim p(\mathrm{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t\mid\boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right].$$

Proof. A sufficient and necessary condition for a vector field to generate a probability path is the *continuity equation*, which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t}p(\boldsymbol{x}, t) + \mathrm{div}\left(p(\boldsymbol{x}, t)\boldsymbol{v}(\boldsymbol{x}, t)\right) = 0, \tag{11}$$

where $\mathrm{div}(f)$ denotes the divergence of function f. The marginal velocity field generates the marginal density $p(\boldsymbol{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\boldsymbol{x}, t)\,\boldsymbol{v}(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)\,\mathrm{d}\boldsymbol{x}_0, \tag{12}$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\boldsymbol{x}, t)$. Dividing by $p(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)\,\mathrm{d}\boldsymbol{x}_0$ and applying Bayes' rule $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x}) = p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)/p(\boldsymbol{x}, t)$:

$$\boldsymbol{v}(\boldsymbol{x}, t) = \int \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x})\,\mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\mathbf{x}_0\mid\mathbf{x}_t=\boldsymbol{x}}[\boldsymbol{v}(\boldsymbol{x}, t \mid \mathrm{x}_0)]\,. \tag{13}$$

□

C Proofs of Theorems and Propositions

C.1 Proof of Theorem 2 (Jacobian Variance Amplification)

Theorem 2 (Jacobian Variance Amplification)**.** *The Jacobi factor amplifies the gradient variance of the original MeanFlow loss:*

$$\mathrm{Var}[\nabla_{\boldsymbol{\theta}}\ell_{MF}(\boldsymbol{\theta}) \mid \boldsymbol{x}_t] \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,.$$

Proof. Under the stop-gradient operator, the compound prediction $V_{\boldsymbol{\theta}}^{\mathrm{sg}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. Let $\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\boldsymbol{\theta}} \ell_{\mathrm{MF}} = 2 \left(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}\right)^{\top} \left(\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}} + \boldsymbol{J} \boldsymbol{v}'\right), \tag{14}$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$ is the mean-field residual and $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$ is the Jacobi factor. All three quantities—$\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}$, $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$, and $\boldsymbol{J}$—are deterministic conditioned on $\boldsymbol{x}_t$.

Since $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$ by equation 1, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2 \left(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}\right)^{\top} \boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}. \tag{15}$$

The centered gradient is $\nabla_{\boldsymbol{\theta}} \ell - \mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \boldsymbol{J} \boldsymbol{v}'$, so the conditional variance is:

$$\begin{aligned} \mathrm{Var}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] &= \mathbb{E}\Big[\big\|2(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \boldsymbol{J} \boldsymbol{v}'\big\|^2 \mid \boldsymbol{x}_t\Big] \\ &= 4\,\mathbb{E}\left[(\boldsymbol{v}')^{\top} \boldsymbol{J}^{\top} \underbrace{(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top}}_{\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}} \boldsymbol{J} \boldsymbol{v}' \;\middle|\; \boldsymbol{x}_t \right] \\ &= 4 \,\mathrm{Tr}\big(\boldsymbol{G}_{\boldsymbol{\theta}} \boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big) \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big), \end{aligned} \tag{16}$$

where $\boldsymbol{G}_{\boldsymbol{\theta}} = (\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses $\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. □

C.2 Proof of Theorem 3 (Semi-Gradient Gap)

Theorem 3 (Semi-Gradient Gap)**.** *The gradients of the MeanFlow loss with and without the stop-gradient operator differ by*

$$\underbrace{2(t{-}r)\,\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{x}_t} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \cdot \boldsymbol{r}_{\boldsymbol{\theta}}\right]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\nabla_{\boldsymbol{\theta}} \,\mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)\right]}_{\text{variance-driven gradient difference}}.$$

Proof. The MeanFlow loss conditioned on $\boldsymbol{x}_t$ decomposes as (cf. equation 5):

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\mathrm{MF}}] = \|\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}\|^2 + \mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big). \tag{17}$$

Without stop-gradient. Both terms depend on $\boldsymbol{\theta}$ (through $\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}$), so the full gradient is:

$$\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] = \nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 + \nabla_{\boldsymbol{\theta}} \,\mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big). \tag{18}$$

With stop-gradient. The JVP terms in $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$ and $\boldsymbol{J}$ are treated as constants, so:

$$\nabla_{\boldsymbol{\theta}}^{\mathrm{sg}} \|\boldsymbol{r}^{\mathrm{sg}}\|^2 = 2 \left(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}\right)^{\top} \boldsymbol{r}^{\mathrm{sg}}, \qquad \nabla_{\boldsymbol{\theta}}^{\mathrm{sg}} \,\mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big) = \mathbf{0}. \tag{19}$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned} &\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] - \nabla_{\boldsymbol{\theta}}^{\mathrm{sg}} \mathbb{E}[\ell] \\ &= \underbrace{\nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 - 2 \left(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}\right)^{\top} \boldsymbol{r}^{\mathrm{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\boldsymbol{\theta}} \,\mathrm{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)}_{\text{variance-driven difference}}. \end{aligned} \tag{20}$$

Expanding the first term: $\nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}\|^2 = 2(\nabla_{\boldsymbol{\theta}} \boldsymbol{r}_{\boldsymbol{\theta}})^{\top} \boldsymbol{r}_{\boldsymbol{\theta}}$, and $\nabla_{\boldsymbol{\theta}} \boldsymbol{r}_{\boldsymbol{\theta}}$ includes the derivative of the JVP terms $(t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}})$ w.r.t. $\boldsymbol{\theta}$, yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \cdot \boldsymbol{r}_{\boldsymbol{\theta}}]$. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\boldsymbol{\theta}} \,\mathrm{Tr}(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top})$. □

C.3 Proof of Theorem 4 (Flow Acceleration Expansion)

Theorem 4 (Flow Acceleration Expansion of the Curvature Gap)**.** *Let $\boldsymbol{v}(\boldsymbol{x}, t) \in C^2$. The curvature gap $\Delta(\boldsymbol{x}_t, r, t) := \boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$ admits a Taylor expansion at $(\boldsymbol{x}_t, t)$ along the flow:*

$$\Delta(\boldsymbol{x}_t, r, t) \;=\; -\frac{(t{-}r)}{2} \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \mathcal{R}_2(\boldsymbol{x}_t, r, t),$$

where $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \triangleq \partial_t \boldsymbol{v} + (\nabla_{\boldsymbol{x}} \boldsymbol{v})\, \boldsymbol{v}$ is the material derivative (a.k.a. flow acceleration) and $\mathcal{R}_2 = \mathcal{O}((t{-}r)^2)$.

Proof. Let $\boldsymbol{x}_\tau$ solve the ODE $\frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = \boldsymbol{v}(\boldsymbol{x}_\tau, \tau)$ with terminal condition $\boldsymbol{x}_t = \boldsymbol{x}_t$. The curvature gap $\Delta = \boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$ can be written as:

$$\Delta(\boldsymbol{x}_t, r, t) = \frac{1}{t-r}\int_r^t \boldsymbol{v}(\boldsymbol{x}_\tau, \tau)\,\mathrm{d}\tau - \boldsymbol{v}(\boldsymbol{x}_t, t) = \frac{1}{t-r}\int_r^t [\boldsymbol{v}(\boldsymbol{x}_\tau, \tau) - \boldsymbol{v}(\boldsymbol{x}_t, t)]\,\mathrm{d}\tau. \tag{21}$$

Define $h = \tau - t \in [-(t-r), 0]$. Since $\boldsymbol{v} \in C^2$, we Taylor-expand $\boldsymbol{v}(\boldsymbol{x}_\tau, \tau)$ about $(\boldsymbol{x}_t, t)$ along the ODE trajectory:

$$\boldsymbol{v}(\boldsymbol{x}_\tau, \tau) = \boldsymbol{v}(\boldsymbol{x}_t, t) + \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t)\,h + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t, t)\,h^2 + O(h^3), \tag{22}$$

where $\frac{\mathrm{D}}{\mathrm{D}t}$ denotes the material derivative along the flow. Substituting and integrating over $h \in [-(t-r), 0]$:

$$\begin{aligned}
\Delta &= \frac{1}{t-r}\int_{-(t-r)}^{0}\left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\,h + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\,h^2 + O(h^3)\right]\mathrm{d}h \\
&= \frac{1}{t-r}\left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\cdot\frac{h^2}{2}\bigg|_{-(t-r)}^{0} + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\cdot\frac{h^3}{3}\bigg|_{-(t-r)}^{0} + O\big((t-r)^4\big)\right] \\
&= \frac{1}{t-r}\left[-\frac{(t-r)^2}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \frac{(t-r)^3}{6}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2} + O\big((t-r)^4\big)\right] \\
&= -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \frac{(t-r)^2}{6}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t, t) + \mathcal{R}_3(\boldsymbol{x}_t, r, t),
\end{aligned} \tag{23}$$

where $\mathcal{R}_3 = O((t-r)^3)$ is the remainder. ☐

C.4 Proof of Proposition 5 (Gradient of VaMF Loss)

Proposition 5 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} \;=\; 2\,\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\cdot\tilde{\boldsymbol{r}}^{EMA}\big]\,,$$

where $\tilde{\boldsymbol{r}}^{EMA} := V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is the mean-field residual under the EMA tangent, and no $\boldsymbol{J}\boldsymbol{v}'$ noise term appears.

Proof. The EMA velocity anchor $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ is deterministic given $\boldsymbol{x}_t$, since the EMA parameters $\bar{\boldsymbol{\theta}}$ are fixed during the gradient step. Under stop-gradient, $V_{\boldsymbol{\theta}}^{\mathrm{EMA}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. By the same argument as equation 14, the per-step gradient is:

$$\nabla_{\boldsymbol{\theta}}\ell = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\big(V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}_{\mathrm{cond}}\big)\,. \tag{24}$$

Writing $\boldsymbol{v}_{\mathrm{cond}} = \boldsymbol{v}(\boldsymbol{x}_t, t) + \boldsymbol{v}'$ and noting that $\tilde{\boldsymbol{r}}^{\mathrm{EMA}} := V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is deterministic given $\boldsymbol{x}_t$:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}}\ell] = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\tilde{\boldsymbol{r}}^{\mathrm{EMA}}, \tag{25}$$

since $\mathbb{E}[\boldsymbol{v}' \mid \boldsymbol{x}_t] = \mathbf{0}$. The tower property $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathrm{MF}}^{\mathrm{EMA}} = \mathbb{E}_{r,t,\boldsymbol{x}_t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}}\ell]\big]$ gives the stated result. Crucially, no $\boldsymbol{J}\boldsymbol{v}'$ term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 14 where the stochastic tangent produces the $\boldsymbol{J}\boldsymbol{v}'$ noise. ☐

C.5 Proof of Proposition 6 (Bias-Variance Tradeoff)

Proposition 6 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as*

$$\tilde{\boldsymbol{r}}^{EMA} \;=\; \boldsymbol{r}_{\boldsymbol{\theta}} \;+\; (t{-}r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t))\,,$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual from equation 5. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$ and the FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \mathbf{0}$.

Proof. Expanding $V_{\boldsymbol{\theta}}^{\text{EMA}}$ from equation 8, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\boldsymbol{\theta}}^{\text{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \,. \tag{26}$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\boldsymbol{r}_{\boldsymbol{\theta}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) - \boldsymbol{v}(\boldsymbol{x}_t, t). \tag{27}$$

Subtracting:

$$\begin{aligned}\tilde{\boldsymbol{r}}^{\text{EMA}} &= V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t) \\ &= \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\, \partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)) \,.\end{aligned} \tag{28}$$

At convergence, the true residual $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 9 supervises the boundary condition, driving $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$ and hence $\tilde{\boldsymbol{r}}^{\text{EMA}} \to \mathbf{0}$. ☐

C.6 Proof of Proposition 7 (Target–Tangent Bias Asymmetry)

Proposition 7 (Target–Tangent Bias Asymmetry)**.** *Let $\hat{\boldsymbol{v}} = \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ be a deterministic proxy with bias $\boldsymbol{b} \triangleq \hat{\boldsymbol{v}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$. For the tangent-replacement loss (VaMF), the loss minimum $\boldsymbol{u}^\star$ satisfies $\boldsymbol{u}^\star(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t) = -(t-r)\, \partial_{\boldsymbol{z}}\boldsymbol{u}^\star\, \boldsymbol{b} + \mathcal{O}(\|\boldsymbol{b}\|^2)$. For the target-replacement loss, the loss minimum at $r = t$ satisfies $\boldsymbol{u}^\star(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t) = \boldsymbol{b}$.*

Proof. **Tangent replacement.** Substituting $\hat{\boldsymbol{v}}$ for $\boldsymbol{v}_{\text{cond}}$ only in the JVP tangent (target unchanged) gives the per-sample compound prediction

$$V_{\boldsymbol{\theta}}^{\text{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{v}} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \,,$$

and the loss is $\mathbb{E}\|V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\|^2$. By stationarity, $\mathbb{E}[V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}} \mid \boldsymbol{x}_t] = \mathbf{0}$ at the minimum. Taking conditional expectation and using $\mathbb{E}[\boldsymbol{v}_{\text{cond}} \mid \boldsymbol{x}_t] = \boldsymbol{v}(\boldsymbol{x}_t, t)$,

$$\boldsymbol{u}^\star + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}^\star \cdot \hat{\boldsymbol{v}} + \partial_t \boldsymbol{u}^\star) = \boldsymbol{v}(\boldsymbol{x}_t, t).$$

At the same point, the *true* MeanFlow identity (with the marginal velocity as tangent) reads

$$\boldsymbol{u}^\diamond + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}^\diamond \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}^\diamond) = \boldsymbol{v}(\boldsymbol{x}_t, t),$$

with $\boldsymbol{u}^\diamond(\boldsymbol{x}_t, t, t) = \boldsymbol{v}(\boldsymbol{x}_t, t)$. Subtracting and writing $\boldsymbol{u}^\star = \boldsymbol{u}^\diamond + \Delta$,

$$\Delta + (t-r)\, \partial_{\boldsymbol{z}}\boldsymbol{u}^\star\, (\hat{\boldsymbol{v}} - \boldsymbol{v}) + (t-r)\, \partial_t \Delta + (t-r)\, (\partial_{\boldsymbol{z}}\Delta)\, \boldsymbol{v} = \mathbf{0}.$$

Evaluating at $r = t$ gives $\Delta(\boldsymbol{x}_t, t, t) = \mathbf{0}$ (boundary condition recovered exactly). For $r < t$, the leading term in $\boldsymbol{b} = \hat{\boldsymbol{v}} - \boldsymbol{v}$ gives

$$\Delta(\boldsymbol{x}_t, r, t) = -(t-r)\, \partial_{\boldsymbol{z}}\boldsymbol{u}^\star\, \boldsymbol{b} + \mathcal{O}(\|\boldsymbol{b}\|^2),$$

i.e., the stationary error is $\mathcal{O}((t-r)\, \|\partial_{\boldsymbol{z}}\boldsymbol{u}\|\, \|\boldsymbol{b}\|)$ and vanishes at the boundary $r \to t$.
Target replacement. Substituting $\hat{\boldsymbol{v}}$ for $\boldsymbol{v}_{\text{cond}}$ in the regression target yields the loss $\mathbb{E}\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\, \mathrm{D}\boldsymbol{u}/\mathrm{D}t - \hat{\boldsymbol{v}}\|^2$. By stationarity, the loss minimum satisfies $\mathbb{E}[\boldsymbol{u}^\star + (t-r)\, \mathrm{D}\boldsymbol{u}^\star/\mathrm{D}t - \hat{\boldsymbol{v}} \mid \boldsymbol{x}_t] = \mathbf{0}$. Specializing to $r = t$, the gap term vanishes and the equation collapses to $\boldsymbol{u}^\star(\boldsymbol{x}_t, t, t) = \hat{\boldsymbol{v}} = \boldsymbol{v}(\boldsymbol{x}_t, t) + \boldsymbol{b}$, so the stationary error at the boundary is exactly $\boldsymbol{b}$, independent of (t, r).
Asymmetry. Tangent replacement carries a multiplicative $(t-r)$ factor on the bias and is therefore pinned to zero at the boundary $r = t$ that the FM anchor enforces; target replacement carries no such factor and the bias persists at every t, requiring a boundary regularizer at *every* (r, t) to remove. ☐

C.7 Proof of Theorem 4 (Optimal Control-Variate Coefficient)

Theorem 8 (Optimal control-variate coefficient)**.** *Under the scalar-isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma^2 \boldsymbol{I}_d$ and $\boldsymbol{J} \approx \kappa \boldsymbol{I}_d$, the conditional MSE of $\boldsymbol{g}^{(\beta)}$ in equation 7 is*

$$M(\beta) \;=\; \beta^2(\kappa+1)^2\|\boldsymbol{b}\|^2 \;+\; \sigma^2 d\big((1-\beta)\kappa - \beta\big)^2,$$

with unique minimizer $\beta^\star = \frac{\kappa}{\kappa+1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\boldsymbol{b}\|^2}$ and optimum value $M(\beta^\star) = \sigma^2 d\, \kappa^2\, \|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. For all κ, $\|\boldsymbol{b}\|^2 > 0$, the optimum strictly dominates both $M(0) = \sigma^2\kappa^2 d$ and $M(1) = (\kappa+1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d$.

Proof. **Step 1: bias-variance decomposition.** From equation 7,

$$\boldsymbol{g}^{(\beta)} \;=\; 2\,\boldsymbol{g}^\top\Big[\bar{\boldsymbol{r}} \;+\; \big((1-\beta)\,\boldsymbol{J} - \beta\,\boldsymbol{I}\big)\,\boldsymbol{v}' \;+\; \beta\,(\boldsymbol{J}+\boldsymbol{I})\,\boldsymbol{b}\Big].$$

Using $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$, the conditional mean is $\mathbb{E}[\boldsymbol{g}^{(\beta)} \mid \boldsymbol{x}_t] = 2\boldsymbol{g}^\top(\bar{\boldsymbol{r}} + \beta(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b})$ and the centered noise is $-2\boldsymbol{g}^\top\big((1-\beta)\boldsymbol{J} - \beta\boldsymbol{I}\big)\boldsymbol{v}'$ (relative to the ideal target $2\boldsymbol{g}^\top\bar{\boldsymbol{r}}$). The conditional MSE w.r.t. the ideal target therefore decomposes as

$$\mathrm{MSE} \;=\; \underbrace{4\beta^2\,\big\|\boldsymbol{g}^\top(\boldsymbol{J}+\boldsymbol{I})\boldsymbol{b}\big\|^2}_{\text{bias}^2} \;+\; \underbrace{4\;\mathrm{Tr}\Big(\boldsymbol{g}\boldsymbol{g}^\top\big((1-\beta)\boldsymbol{J} - \beta\boldsymbol{I}\big)\Sigma_{\boldsymbol{v}'}\big((1-\beta)\boldsymbol{J} - \beta\boldsymbol{I}\big)^\top\Big)}_{\text{variance}}.$$

Step 2: scalar approximation. Substituting $\boldsymbol{J} = \kappa\boldsymbol{I}_d$ and $\Sigma_{\boldsymbol{v}'} = \sigma^2\boldsymbol{I}_d$, and absorbing the parameter-direction normalization $\mathrm{Tr}(\boldsymbol{g}\boldsymbol{g}^\top) = \|\boldsymbol{g}\|_F^2$ into a per-component MSE,

$$M(\beta) \;\propto\; \beta^2(\kappa+1)^2\|\boldsymbol{b}\|^2 \;+\; \sigma^2 d\big((1-\beta)\kappa - \beta\big)^2.$$

Step 3: minimization. Differentiating and setting to zero,

$$\begin{aligned}\frac{\partial M}{\partial\beta} &= 2\beta(\kappa+1)^2\|\boldsymbol{b}\|^2 - 2(\kappa+1)\sigma^2 d\big((1-\beta)\kappa - \beta\big) \;=\; 0\\ \iff\quad & \beta(\kappa+1)\big(\sigma^2 d + \|\boldsymbol{b}\|^2\big) \;=\; \sigma^2 d\,\kappa.\end{aligned}$$

Solving yields $\beta^\star = \sigma^2 d\,\kappa/[(\kappa+1)(\sigma^2 d + \|\boldsymbol{b}\|^2)]$, which factors as the noise-cancellation $\kappa/(\kappa+1)$ times the James-Stein shrinkage $\sigma^2 d/(\sigma^2 d + \|\boldsymbol{b}\|^2)$.

Step 4: optimum value. At $\beta^\star$, $(1-\beta^\star)\kappa - \beta^\star = \kappa\,\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. Substituting back,

$$M(\beta^\star) \;=\; \frac{\sigma^4 d^2\kappa^2\|\boldsymbol{b}\|^2 + \sigma^2 d\,\kappa^2\|\boldsymbol{b}\|^4}{(\sigma^2 d + \|\boldsymbol{b}\|^2)^2} \;=\; \frac{\sigma^2 d\,\kappa^2\,\|\boldsymbol{b}\|^2}{\sigma^2 d + \|\boldsymbol{b}\|^2}.$$

Step 5: strict dominance over corners. For $\kappa, \|\boldsymbol{b}\|^2 > 0$: $M(\beta^\star)/M(0) = \|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2) < 1$, so $M(\beta^\star) < M(0)$. For the $M(1)$ comparison, $M(1) - M(\beta^\star) = (\kappa+1)^2\|\boldsymbol{b}\|^2 + \sigma^2 d - \sigma^2 d\kappa^2\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$. The first two terms exceed $\sigma^2 d\kappa^2\|\boldsymbol{b}\|^2/(\sigma^2 d + \|\boldsymbol{b}\|^2)$ since $(\kappa+1)^2 > \kappa^2$ for $\kappa > 0$. ☐

C.8 Derivation of the Trace Weight

The control-variate optimum of Theorem 4 suggests an alternative *loss-side* reweighting: rather than tuning the tangent coefficient β, equalize per-sample loss magnitudes by downweighting high-variance samples directly. We formalize this as a *trace weight*

$$w(\boldsymbol{x}_t, r, t) \;:=\; \frac{1}{1 + \sigma_t^2\,\mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]\,/d} \;=\; \frac{d}{d + \sigma_t^2\,\mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]}, \tag{29}$$

derived below as a regularized BLUE-scalar approximation under the isotropic-noise model. Empirically, the trace weight provides marginal additional gain once the EMA tangent removes the dominant Jacobian amplification, so we keep this construction as an appendix-only Lagrangian implementation of the curvature-budget constraint.

Setup. Under the EMA tangent (Section 3.3), proposition 5 and the Reynolds decomposition imply that the per-sample loss conditioned on $\boldsymbol{x}_t$ admits the noise-decomposition

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\Big[\big\|V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}_{\mathrm{cond}}\big\|_2^2\Big] \;=\; \underbrace{\big\|\tilde{\boldsymbol{r}}^{\mathrm{EMA}}\big\|_2^2}_{\text{deterministic signal}} \;+\; \underbrace{\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)}_{\text{irreducible noise floor }\nu(\boldsymbol{x}_t,r,t)}. \tag{30}$$

The noise floor ν is heteroscedastic: it varies with $(\boldsymbol{x}_t, r, t)$ through both $\boldsymbol{J}$ and $\Sigma_{\boldsymbol{v}'}$, and dominates the per-sample loss in high-curvature regions even though it does *not* contribute to the parameter gradient (its $\boldsymbol{\theta}$-dependence is wiped by the stop-gradient on the JVP).

Heteroscedastic-noise BLUE. Suppose we form a weighted empirical-risk estimator $\hat{\mathcal{L}}(\boldsymbol{\theta}) = \frac{1}{N}\sum_i w_i\,\ell_i(\boldsymbol{\theta})$ from N independent per-sample losses ℓ_i whose noise floors $\nu_i = \mathrm{Var}[\ell_i \mid \boldsymbol{\theta}]$ are heteroscedastic and known. The Best Linear Unbiased Estimator (BLUE) of the population mean assigns

$$w_i^{\mathrm{BLUE}} \;\propto\; \nu_i^{-1}, \tag{31}$$

minimizing the variance of $\hat{\mathcal{L}}$ subject to $\mathbb{E}[\hat{\mathcal{L}}] = \mathbb{E}_{\boldsymbol{x}_t,r,t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell]\big]$. Applying equation 31 to our setting with $\nu_i = \mathrm{Tr}(\boldsymbol{J}_i\Sigma_{\boldsymbol{v}',i}\boldsymbol{J}_i^\top)$ would prescribe $w \propto 1/\,\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$. Two practical obstacles remain:

1. The matrix BLUE $w \propto (\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)^{-1}$ at the gradient level (rather than the trace) requires a per-sample $d\times d$ inversion, costing $\mathcal{O}(d^3)$ per sample.
2. The scalar form $w \propto 1/\,\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ is unbounded as $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \to 0$ (low-curvature regions), where the BLUE prescribes infinite weight on samples with vanishing noise floor.

Isotropic approximation and Tikhonov stabilization. We resolve both pathologies in two steps. First, the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$ collapses the trace to a scalar:

$$\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) \;=\; \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top), \tag{32}$$

which is a single Hutchinson estimate at $\mathcal{O}(d)$ cost. (Appendix D.2 discusses concrete choices of σ_t.) Second, we add a Tikhonov regularizer of magnitude d to the denominator:

$$w(\boldsymbol{x}_t, r, t) \;:=\; \frac{d}{d + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;=\; \frac{1}{1 + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d}, \tag{33}$$

matching equation 29. The regularization has three desirable properties:

- **Boundedness.** $w \in (0,1]$ for all $(\boldsymbol{x}_t, r, t)$, with $w = 1$ exactly when $\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = 0$ (no reweighting in flat regions).
- **BLUE limit.** As $\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \to \infty$, $w \to d/[\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]$, recovering the BLUE-scalar prescription equation 31 up to a constant.
- **Bias-variance tradeoff.** The added d trades a small estimator bias (the BLUE limit is approached but not attained) for bounded weights, eliminating the divergence at low curvature.

The choice of regularizer magnitude d has a clean physical reading: under the BLUE limit the noise contribution per sample equals $w \cdot \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = d$ (i.e., one unit per data dimension), and the regularizer is the smallest constant for which $w \le 1$ uniformly.

C.9 Proof of Proposition 9 (Trace-Weight Properties)

Proposition 9 (Trace-Weight Properties)**.** *Under the trace-weight loss equation 10 and the isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$: **(a)** the mean-parameter gradient is the trace-weighted form of proposition 5 with w as a stop-gradient scalar, so the BLUE-style normalization does not enter the parameter update; **(b)** the expected per-sample contribution to the loss decomposes into a deterministic signal $w\,\|\tilde{\boldsymbol{r}}^{EMA}\|_2^2$ and a constant noise floor $w\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/(d + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)) \le d$, eliminating the unbounded curvature growth from theorem 2; **(c)** w is monotonically decreasing in $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ and, by Theorem 4, in $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$, so high-curvature long-interval samples are automatically downweighted.*

Proof. **(a) Gradient form.** The trace weight w in equation 29 is wrapped in a stop-gradient operator on $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, so $\nabla_{\boldsymbol{\theta}} w = 0$ and w acts as a fixed scalar multiplier. Differentiating the trace-weighted MSE in equation 10 therefore gives

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathrm{VaMF}}^{\mathrm{MF}} \;=\; 2\,\mathbb{E}\big[w\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\tilde{\boldsymbol{r}}^{\mathrm{EMA}}\big], \tag{34}$$

identical to proposition 5 except that each per-sample contribution carries the scalar weight w. The conditional fluctuation $\boldsymbol{v}'$ produces a cross term $\mathbb{E}[(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top \boldsymbol{v}'] = \boldsymbol{0}$ that vanishes by $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$, so the BLUE-style normalization does not propagate into the gradient computation.

(b) Bounded noise floor. Writing $V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}} = \tilde{\boldsymbol{r}}^{\text{EMA}} - \boldsymbol{v}'$ with the EMA tangent making $\tilde{\boldsymbol{r}}^{\text{EMA}}$ deterministic given $\boldsymbol{x}_t$, the inner expectation of the per-sample loss is

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[w\,\|V_{\boldsymbol{\theta}}^{\text{EMA}} - \boldsymbol{v}_{\text{cond}}\|_2^2\big] \;=\; w\,\|\tilde{\boldsymbol{r}}^{\text{EMA}}\|_2^2 + w\cdot\sigma_t^2\,\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}), \tag{35}$$

where the cross term vanishes as in (a). Substituting the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$ and the explicit form of w yields the noise contribution

$$w\cdot\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \;=\; \frac{\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{1+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d} \;=\; \frac{d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{d+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;\le\; d, \tag{36}$$

with equality only in the limit $\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \to \infty$. The trace weight thus caps the noise term at the data dimensionality, eliminating the unbounded curvature growth from theorem 2.

(c) Curvature-aware downweighting. The map $w(\xi) = 1/(1+\sigma_t^2\xi/d)$ is strictly decreasing in $\xi = \mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \ge 0$. By Theorem 4 and the Reynolds decomposition, $\mathbb{E}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]$ scales with $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$ to leading order, so w shrinks on the same factor and high-curvature long-interval samples are automatically downweighted—realizing the curvature-aware schedule prescribed in Appendix C.8 in a single per-sample scalar without requiring an explicit progressive schedule or interval-importance sampling. ☐

D Additional Results

D.1 Acceleration-to-Noise Ratio from the Trace Weight

For completeness we record an alternative interpretation of the trace weight in proposition 9 as an empirical proxy for the squared *acceleration-to-noise ratio*. This characterization predates the control-variate framing of Theorem 4 but remains a useful sanity check in the small-curvature-gap regime; the connection becomes precise when the schedule σ_t is chosen consistently with the actual conditional-velocity covariance.

Proposition 10 (ANR from the Trace Weight)**.** *Let $\rho(\boldsymbol{x}_t, r, t) := \sigma_t^2\ \mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$ denote the curvature term in the denominator of equation 29, and assume the schedule satisfies $\sigma_t^2 = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$ (the moment-matched isotropic approximation, achieved exactly by the* BLUE-GAUSS *schedule of Appendix D.2 under Laplace's approximation of p_{data}). Then near convergence,*

$$\rho \;\approx\; \frac{\|\Delta(\boldsymbol{x}_t, r, t)\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})} \;\approx\; \frac{(t-r)^2}{4}\cdot\frac{\big\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t,t)\big\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})} \;=\; \frac{\mathrm{ANR}(\boldsymbol{x}_t,r,t)^2}{4}, \tag{37}$$

where $\mathrm{ANR} := (t-r)\big\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\big\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$. The trace weight is then $w = 1/(1+\rho) = 1/(1+\mathrm{ANR}^2/4)$.

Proof. **Step 1: noise floor and curvature gap.** Under the EMA tangent (Section 3.3) and at convergence, proposition 6 gives $\tilde{\boldsymbol{r}}^{\text{EMA}} \approx \boldsymbol{0}$, so the per-sample loss reduces to its noise floor $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ (cf. equation 30). Substituting the Reynolds-residual expansion $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} = \Delta + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}_{\boldsymbol{\theta}}$ from Section 3.1 and using the MeanFlow identity $(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u} = -\Delta$ gives the per-sample noise floor $\big\|\boldsymbol{J}\boldsymbol{v}'\big\|_2^2$ at first order; taking expectations and applying the trace identity yields

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[\|\boldsymbol{J}\boldsymbol{v}'\|_2^2\big] \;=\; \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big) \;\approx\; \big\|\Delta(\boldsymbol{x}_t,r,t)\big\|^2 \tag{38}$$

in the small-gap regime.

Step 2: isotropic moment matching. Under the schedule assumption $\sigma_t^2 = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$, the isotropic-noise approximation equation 32 gives $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) = \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$. Combining with equation 38,

$$\frac{\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{d} \;\approx\; \frac{\|\Delta\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})} \quad\Longleftrightarrow\quad \rho \;\approx\; \frac{\|\Delta\|^2}{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}. \tag{39}$$

Step 3: substitute the flow-acceleration expansion. Theorem 4 gives $\|\Delta\| \approx \frac{t-r}{2}\big\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\big\|$ to leading order, so $\|\Delta\|^2/\,\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}) \approx (t-r)^2\big\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\big\|^2/[4\ \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})] = \mathrm{ANR}^2/4$, completing equation 37. ☐

Schedule sensitivity. The proof requires the moment-matched schedule $\sigma_t^2 = \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$. Practical schedules (Appendix D.2) only approximate this:

- BLUE-GAUSS matches it exactly under Laplace's approximation of p_{data}, so equation 37 holds to leading order.
- U-SHAPE qualitatively tracks the location of the peak of $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ (eq. (41)) but not its absolute scale, so the connection holds up to a t-dependent multiplicative constant.
- LINEAR and QUADRATIC are monotone in t and *do not* approximate the U-shape; the connection in equation 37 is then valid only as a qualitative trend rather than a quantitative identity.

The trace weight remains a valid BLUE-scalar approximation in all three cases (Appendix C.8); only the ANR *interpretation* of ρ is schedule-sensitive.

D.2 Schedule σ_t in the Trace Weight

The trace weight equation 29 introduces a single hyperparameter, the time-dependent scalar σ_t that approximates the conditional-velocity standard-deviation scale via $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$. The exact $\Sigma_{\boldsymbol{v}'}$ depends on p_{data} and is not generally closed-form; we first derive a tractable instance under Laplace's approximation $p_{\text{data}} \approx \mathcal{N}(\boldsymbol{0}, \sigma_d^2 \boldsymbol{I}_d)$, then compare five practical schedules empirically.

Conditional-velocity variance under Laplace's approximation. Following Laplace's approximation in Bayesian statistics [38, 39], we replace p_{data} by a Gaussian centered at the mode with covariance set to the inverse Hessian of $-\log p_{\text{data}}$ at that mode. Take $\boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{0}, \sigma_d^2 \boldsymbol{I}_d)$ and $\boldsymbol{x}_1 \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$ with the linear interpolant $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$. Conditional on $\boldsymbol{x}_0$, the noise contribution is $\boldsymbol{x}_t \mid \boldsymbol{x}_0 \sim \mathcal{N}((1-t)\boldsymbol{x}_0, t^2 \boldsymbol{I}_d)$. Combining this Gaussian likelihood with the Gaussian prior on $\boldsymbol{x}_0$ via conjugate inference,

$$\mathrm{Var}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t] \;=\; \frac{t^2 \sigma_d^2}{\sigma_d^2 (1-t)^2 + t^2}\, \boldsymbol{I}_d. \tag{40}$$

Substituting $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0 = (\boldsymbol{x}_t - \boldsymbol{x}_0)/t$ gives

$$\Sigma_{\boldsymbol{v}'}(\boldsymbol{x}_t, t) \;=\; \frac{1}{t^2}\,\mathrm{Var}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t] \;=\; \frac{\sigma_d^2}{\sigma_d^2 (1-t)^2 + t^2}\, \boldsymbol{I}_d. \tag{41}$$

Equation (41) is U-shaped in t: at $t = 0$ and $t = 1$ it equals σ_d^2 (resp. 1 for the unit-variance case $\sigma_d = 1$), and at $t = 0.5$ it reaches its maximum of $2\sigma_d^2/(\sigma_d^2 + 1)$. The peak coincides exactly with the regime where the Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ from theorem 2 is largest, since both reflect the same physical phenomenon—mid-t mode mixing.

Practical schedules. Real p_{data} (e.g. image latents) is non-Gaussian and Eq. equation 41 is at best a first-order approximation. We therefore benchmark five practical schedules:

NONE. $\sigma_t = 1$. The bare BLUE-scalar form. Serves as the cleanest test of the curvature term in isolation.

LINEAR. $\sigma_t = t$. The simplest monotone schedule with $\sigma_t^2 = t^2$ in the denominator of equation 29. Suppresses high-t samples where the noise has dispersed.

QUADRATIC. $\sigma_t = t^2$, so $\sigma_t^2 = t^4$. Steeper monotone profile, used in early experiments.

U-SHAPE. $\sigma_t = \sqrt{t(1-t)}$, so $\sigma_t^2 = t(1-t)$. Peaks at $t = 0.5$ and vanishes at the endpoints—qualitatively matching the location of equation 41's peak without committing to a parametric form.

BLUE-GAUSS. $\sigma_t^2 = 1/(\sigma_d^2(1-t)^2 + t^2)$ from equation 41 with $\sigma_d = 1$. The closed-form BLUE under Laplace's approximation of p_{data}.

The toy benchmark sweep across all five schedules appears in Section 5, the toy schedule sweep (Appendix D.2); the empirical winner ($\sigma_t = 1$) is used as the default in the main text and in Appendix E.

E DiT Latent-Space Training on ImageNet

We test whether the variance-reduction mechanism of Theorems 2 and 4 scales beyond toys by training DiT [40] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, following the recipe of [20]. Both baselines and VaMF runs share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE versus the trace-weighted MSE in equation 10 with $\sigma_t = 1$, the toy-sweep winner of the toy schedule sweep (Appendix D.2), and a single Hutchinson probe).

E.1 DiT-B/4, ImageNet-256×256

Setup: logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with constant learning rate 10^{-4}, EMA decay 0.9999, batch size 256 across 32 TPU v4 chips, $300{,}000$ steps. The single-probe Hutchinson estimator adds approximately one additional JVP per step, which we measure as a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for VaMF-TW versus 2.65 steps/s for the baseline). At identical optimizer step counts, VaMF-TW maintains a small but consistently lower per-sample loss than the baseline beyond $\sim 20{,}000$ steps. The matched-step loss comparison in Table 5 (baseline run `fidpdet7` and VaMF-TW run `ymxp1xfg` on Weights&Biases) reflects the latest snapshot before submission; the VaMF-TW run is on track to complete the full $300{,}000$ steps shortly after the abstract deadline, at which point we will add the $\text{FID}_{50\text{k}}$ comparison to the camera-ready version.

Table 5: Matched-step training-loss comparison on the DiT-B/4 ImageNet-256 latent run. Each cell averages the per-step training MSE over the 30 logged points immediately preceding the listed step. Negative deltas indicate VaMF (with EMA proxy + FM anchor + trace-weight reweighting per Appendix C.8) achieves a lower training loss than the baseline beyond step $\sim$ 20k, providing scale-up evidence consistent with the toy-benchmark variance reduction of Figure 2.

Step	MeanFlow (`fidpdet7`)	VaMF-TW (`ymxp1xfg`)	Δ
$10{,}000$	$3{,}718$	$3{,}736$	$+0.5\%$
$20{,}000$	$3{,}827$	$3{,}759$	-1.8%
$30{,}000$	$3{,}888$	$3{,}884$	-0.1%
$35{,}500$	$3{,}953$	$3{,}885$	-1.7%